

* Assignment - 2 [ch. - 1, 2, 3] *

Q.-1 Answer the following Question. Carry 3

② Explain the OSI security architecture in brief.

Ans.- It provides a ~~fundamental~~ foundation framework for secure communication defined by the ISO standard.

- It offers a systematic approach to understand and implement network security, guiding professionals in addressing diverse threats within complex digital environments.
- OSI [Open security interconnection] security refers to a set of protocols standards and techniques used to ensure the security of data and communicates in a network environment based on the OSI model.
- Some concepts are:

1) security attack

- It is an attempt by a person/entity to gain unauthorized access to disrupt the security of a system, network or device.

2) security mechanism

- The mechanism that is ~~used~~ built to identify any breach of security or attack on organization is called a security mechanism. It is responsible for protecting a system, network or device against unauthorized access, tampering or other security threats.

3) security services

- It refers to the different services available for maintaining the security and safety of an organization. They help in preventing any potential risk to security.

- security across OSI layers.

* Layer 2 [Data Link]

~~MACsec~~ MACsec - provide hop by hop encryption for ethernet frames, securing communication between directly connected devices.

ex.- protecting sensitive data flow between switches in a data center.

* Layer 3 [Network]

IPsec - A suite of protocols used to secure IP communicates by authenticating

and encrypting each IP packet.
ex.- creating a secure tunnel for remote employees to access corporate resources.

Layer 4 [Transport]

TLS/SSL - provides end to end encryption and authentication between server and client for appli. ~~data~~ level data.

ex.- securing web browsing (HTTPS), email (SMTPS) and etc.

Layer 7 [Application]

SSH - Enables secure remote access to computers over an unsecured network.

ex.- encrypting sensitive emails.

- Benefits:

1) Interoperability

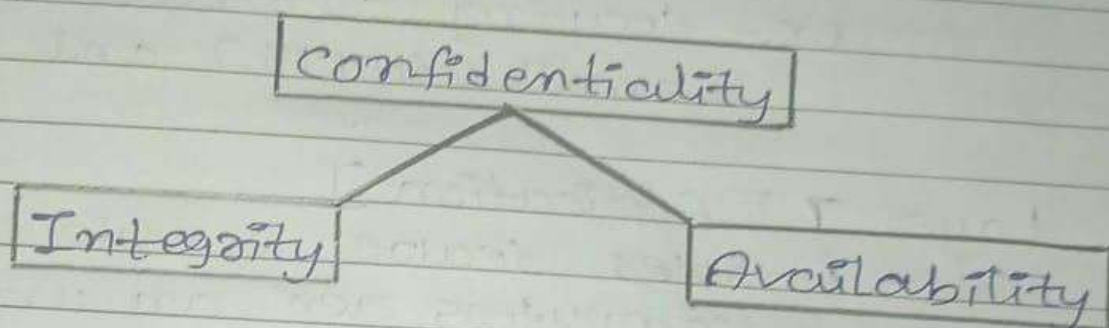
- The OSI model divides network functions into multiple levels makes it easier for different hardware and software components to work together.

2) Meets International Standards

- security services are defined and recognized internationally meeting international standards.

③ Explain the CIA triad in detail.
Ans- CIA stands for Confidentiality, Integrity and Availability.

- It is a foundational model in IS. It serves as a guiding framework for policies and practices that aim to protect critical information and ensure the reliability of systems within an organization.



- CIA triad:

1) Confidentiality

- It ensures that sensitive information is accessible only to authorized systems or individuals and prevent unauthorized access. The goal is to protect private data from being viewed, accessed or used by unauthorized persons.

ex:- If anyone has seen password of your gmail account or of any other platform, in that case your password has been compromised &

confidentiality has been breached.

2) Integrity

- It ensures that data remains unaltered during transmission ~~for~~ ~~stage~~ storage. If the data is modified in any way, its integrity is compromised. When data is corrupted it means the integrity is lost, leading to potential errors or malicious changes.

ex.- If an employee left the organization then all data of employee like account and other information should be updated to reflect status as 'JOB LEFT', so that data is complete and accurate and in addition to this, only authorized person should allow to edit employee data.

3) Availability

- It ensures that information must be available when needed. A network that is unavailable can disrupt business operations, causing significant issues for companies and users relying on it.

ex.-

a) Redundant System

- Deploying multiple servers to ensure that if one system fails, another can take over, maintaining service continuity.

b) Load balancing

- Distributing network traffic across multiple servers to prevent single server from being overburden or unavailability.
-

④ List all the security threats & attacks. Explain any 3 in detail.

Ans. - ~~Security~~ Security threats and attacks in information. Security ~~compromises~~ encompass a wide range of malicious activities aimed at compromising the CIA of data and system.

- Security attacks are:

- 1) Malware attacks
- 2) Ransomware attacks
- 3) Phishing attacks
- 4) Social engineering attacks
- 5) Denial-of-services (DoS / DDoS)

- security threats are:

- 1) AI-Powered threats
- 2) cloud ~~or~~ security ~~challenges~~ challenges
- 3) supply chain attacks
- 4) Ransomware surge

- explanation:

1) AI-Powered threats

- AI and machine learning are increasingly used by attackers to create sophisticated phishing campaigns and generate advanced malware, making them harder to detect. Attackers use AI to automate the process of finding ~~the~~ ~~in~~ vulnerabilities, gathering information about targets and planning attack strategies, allowing them to launch attacks at scale and speed.

2) Malware attacks

- Malware is malicious software and refers to any software that is designed to cause harm to computer systems, networks or users. It includes computer viruses, worms, Trojan horses, Ransomware, spyware and other malicious programs.

3) Denial of Services (DoS)

- It is a cyberattack that aims to make a computer, network, or online services unavailable to its intended users by overwhelming it with a massive flood of illegitimate traffic or requests. The attackers send a large volume of data or connection requests to the target system.
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Q.-2 Answer the following question. (any 5)

(1) Explain Transposition Techniques and its types with example.

Ans:- It is an encryption method used to encrypt a message or information. This encryption method is done by playing with the position of letters of the plain text. The positions of characters present in the plaintext are rearranged to form the ciphertext.

- Types:

1) Rail Fence transposition cipher

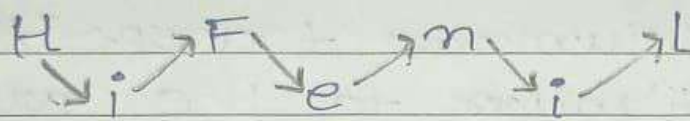
- It is simplest transposition cipher techniques. It is also termed as a zigzag cipher. It gets its name from



the way through which it performs encryption of plain-text.

-Technique:

ex.- The plain text is "Hi Fenil", now we will write this plain text in diagonal form.



ciphertext = HFNLIEI

2) Block (Single Columnar) transposition cipher

- It is used to encrypt the message. In this technique, we first write message in rows, after that we read the message column by column.

-Technique:

ex.- "Transposition" is a plaintext, now we will write the plaintext in the form of row and column.

data = Transposition

key = Cipher
145326

1	2	3	4	5	6
T	r	a	n	s	p
o	s	i	t	i	o
n					

= ton nt si ai rs po

3) Double columnar transposition cipher

- It is similar to the columnar transposition technique. The main objective of using a double columnar is to encrypt the message twice.
- It makes use of single columnar technique but uses two times, it can use same or different secret keys.
- The output obtain from first encryption will be the input to the second encryption.

② Describe steganography in brief.

- Ans.- It is a practice of concealing information within another message to avoid detection, used to hide virtually any type of digital content, text, image, audio, video, content.
- It is relevant to cybersecurity because some gangs and other threat

actors often hide information when attacking targets.

- Types:

1) Text Steganography

- It involves hiding information inside text files. This includes changing the format of existing text, changing words within a text, using context-free grammar to generate readable texts.

2) Image Steganography

- It involves hiding information within image files. In digital steganography, images are often used to conceal information because there are large number of elements within the digital representation of image, and there are various way to hide information inside an image.

3) Audio Steganography

- It involves secret messages being embedded into an audio signal which alters the binary sequence of the corresponding audio file. Hiding secret messages

in digital sound is more difficult process compared to others.

4) Video steganography

- This is where data is concealed within digital video formats. It allocates large amount of data to be hidden within a moving stream of images and sounds. Two types are - embedding data in uncompressed raw video and then compressing it later & embedding data directly into the compressed data stream.

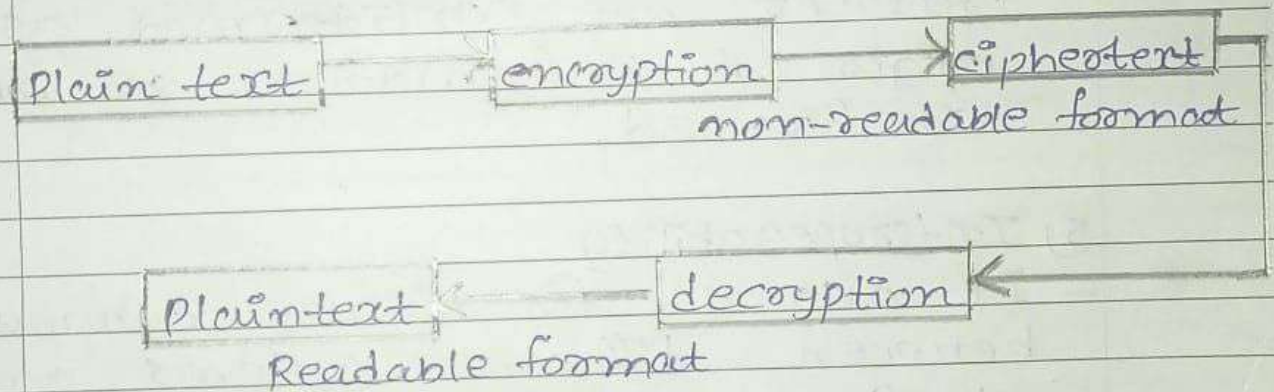
5) Network steganography

- It sometimes known as protocol steganography, a technique of embedding information within network control protocols used in data transmission such as TCP, UDP, ICMP and etc.

④ Discuss features of cryptography.

Ans. - It is a technique of securing information and communications through the use of codes so that only those persons for whom the information is intended can understand and process it.

- Thus, preventing unauthorized access to information.
- The 'crypt' means 'hidden' & 'graphy' means 'writing'.
- Algorithms are used for cryptographic key generation, digital signing and verification to protect data privacy, web browsing on the internet and to protect confidential transactions like credit card and debit card transactions.



- features:

1) Confidentiality

- Information can only be accessed by the person for whom it is intended and no other person except him can access it.

2) Integrity

- Information can't be modified in

Storage or transition between sender and intended receiver without any addition to information being detected.

3) Non-repudiation

- The sender of information can't deny his intention to send information at a later stage.

4) Authentication

- The identities of sender and receiver are confirmed, as well as origin of information ~~and~~ is confirmed.

5) Interoperability

- It allows for secure communication between different systems and platforms.

6) Adaptability

- It continuously evolves to stay ahead of security threats and technological advancements.

⑤ Encrypt given data using playfair cipher method.

data = mechanisms

key = cipher

Ans.- Data = mechanisms
Key = cipher

c	i	p	h	e
ø	a	b	d	f
g	k	L	m	n
o	q	s	t	u
v	w	x	y	z

- encrypted data:

me	ch	an	is	ms
mø	h	ie	fk	l+

⑥ Perform encryption on given data using Vigenere cipher technique.
Data = Substitution technique
Key = Sequence

Ans.-

1	a	9	i	17	q	25	y
2	b	10	j	18	ø	26	z
3	c	11	k	19	s		
4	d	12	L	20	t		
5	e	13	m	21	u		
6	f	14	n	22	v		
7	g	15	o	23	w		
8	h	16	p	24	x		

- Data = substitution technique
Key = sequence

- step 1:

s	u	b	s	t	i	t	u	t	i	o	n	t	e	c	h	n	i	q	u	e
s	e	q	u	e	n	c	e	s	e	q	u	e	n	c	e	s	e	q	u	e

- step 2:

19	21	2	19	20	9	20	21	20	9	15	14	20	5	3	8	14	9	17	21	5
19	5	17	21	5	14	3	5	19	5	17	21	5	14	3	5	19	5	17	21	5

- step 3: $C P + K$

38	26	19	40	25	23	23	26	39	14	32	35	25	19	6	13	33	14	34	42	10
----	----	----	----	----	----	----	----	----	----	----	----	----	----	---	----	----	----	----	----	----

- step 4: mod 26 (if value is greater than 26)

12	²⁶ 0	19	14	25	23	23	26	13	14	6	9	25	19	6	13	7	14	8	16	10
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- step 5: ~~convert~~ convert alphabet from number

L	Z	S	N	Y	W	W	Z	M	N	F	I	Y	S	F	M	G	N	H	P	J
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- So, encrypted data is:

LZSNYWWMNFIFYSEFMGNHPJ

Q.-3 Answer the following question. (any 3)

(1) Explain Block cipher and its design principles.

Ans.- Block ciphers are built in the feistel cipher structure. it has specific no. of rounds and keys for generating ciphertext.

- It is a type of encryption algorithm that processes fixed-size blocks of data, usually 64 or 128 bits, to produce ciphertext.

- Block cipher principle:

1) number of rounds

- No. of rounds are regularly ~~considered~~ considered in design criteria, it just reflects the no. of rounds to be suitable for an algorithm to make it more complex.

2) Design of function F

- The core part of the feistel block cipher structure is the round function. The complexity of cryptanalysis can be derived from round function. i.e., the increasing level of complexity for the round function could be greatly contributing to an increase in complexity.

3) Key size

- It should be large enough to prevent brute-force attacks, large key size means that there are more possible keys, making it harder for an attacker to guess the correct one. A key size of 128 bits is considered to be secure for most apps.

4) Key schedule

- It should be designed carefully to ensure that the keys used for encryption are independent and unpredictable. The key schedule should also resist attacks that exploit weak keys/key-dependent properties of cipher.

5) Block size

- It should be large enough to prevent attack that exploit statistical patterns in the plaintext. A block size of 128 bits is generally considered to be secure for most apps.

6) Avalanche effect

- The cipher should exhibit the avalanche effect, which means that a small change in the plaintext or key should cause a signi-

significant change in the ciphertext. This ensures that any change in the input results in a complete change in output.

7) Security analysis

8) Confusion & Diffusion

② Discuss block cipher modes of operation.

Ans.- It defines how to securely encrypt and decrypt large amounts of data using a block cipher.

- A block cipher is an encryption algorithm that processes data in fixed-size blocks (128 bits) rather than one bit at a time.
- However, to encrypt data larger than a single block, different modes of operation are used to ensure both security and efficiency, here are few common modes,

1) Electronic code book [ECB]

- It is the easiest block cipher mode of functioning. It is easier because of the direct encryption of each block of input plaintext & output

is in the form of blocks of encrypted ciphertext.

- advantages:

- i) Parallel encryption of blocks of bits is possible, thus it is a faster way of encryption.
- ii) Simplest way of block cipher.

- disadvantages:

- i) Prone to cryptanalysis since there is a direct relationship between plaintext and ciphertext.

2) Cipher block chaining [CBC]

- It is an advancement made on ECB since ECB compromises some security requirements.

- In CBC, the previous cipher block is given as an input to the next encryption algorithm after XOR with the original plaintext block.

- advantages:

- i) More secure than ECB because it hides pattern.

- disadvantages:

- i) Parallel encryption is not possible since every encryption requires a previous cipher.

3) Cipher feedback mode [CFM]

- In this, the cipher is given as a feedback to the next block of encryption with some new specifications.
- first, an initial vector IV is used for first encryption & output bits are divided as a set of s & $b-s$ bits.
- The left hand side s bits are selected along with plaintext bits to which an XOR operation is applied.
- The result is given as input to a shift register having $b-s$ bits to lfs ~~lfs~~, bits to rhs and the process continues.
- advantages:
 - i) since, there is some data loss due to use of shift register, thus it is difficult for applying cryptanalysis.
- disadvantages:
 - i) slightly more complex and can propagate errors.

4) Output feedback mode [OFM]

- OFM follows nearly same process as the CFM except that it

sends the encrypted output as feedback instead of actual cipher which is XOR cipher.

- Here all bits of the block are sent instead of sending selected s bits.

- advantages:

i) In the case of CFB, a single bit error in a block is propagated to all subsequent blocks.

This problem is solved by OFB as it is free from bit errors in the plaintext block.

- disadvantages:

i) The drawback of OFB is that, because to its operational modes, it is more susceptible to a message stream modification attack than CFB.

5) Counter mode [CTR]

④ List MITM attacks and explain any 3 in detail.

Ans.- MITM attacks are:

- 1) wifi eavesdropping
- 2) DNS spoofing
- 3) IP spoofing

- 4) HTTPS spoofing
- 5) ARP spoofing
- 6) E-mail spoofing
- 7) Session hacking
- 8) SSL stripping
- 9) MITM attack

- explanation:

1) DNS Spoofing

- DNS cache poisoning, is an attack involving manipulation DNS records to redirect users towards a fraudulent, malicious websites that may resemble the user's intended destination.
- for ex., a server is used by several sites to interpret the address to a recognizable title - google.com. A DNS server transforms 192.156.65.118 to google.com.
- The fraudulent server may transport user requesting google.com webpage from 192.156.56.118 web address to another IP address like 192.146.15.18.
- Attacker can modify an authorized DNS server's records, redirecting traffic to the fraudulent website.

2) HTTPS spoofing

- It is a cyberattack where attackers create false/fraudulent websites that closely mimic the original ones.
- These false websites often display "http://" in address bar, tricking users into believing that they are on a secure sites.
- By registering domain names that look similar to the target website and obtaining SSL certificates, attackers can make their spoofed sites appear original.
- This traffic is also called - "Homograph attack".
- ex. Rolex is written as Rólex.

3) MITB attack

- This is a form of attack that leverages internet browser security flaws. The malicious attacks will be trojans, desktop worms, Java vulnerabilities, SQL injection attacks and webbrowser add-ons. These are commonly used to collect financial information.
- Malware steals bank passwords during login. It may also transfer money and then alter the receipt to hide the transaction.